

Mathematical Foundations for Machine Learning

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1. Solution of Linear Systems

Linear systems are one of the first places where machine learning becomes matrix mathematics. A dataset can place one observation in each row, a model parameter in each column, and a target value on the right. The compact equation $Ax = b$ then asks whether one parameter vector x can satisfy all observations at once.

This note covers the lecture-supported part of Handout Topic 1: matrices, row operations, echelon forms, rank, consistency, and the possible solution types. [Detailed External Sources](#)

From Equations to a Matrix

Consider the equations

$$\begin{aligned}x + 2y &= 5 \\ 3x - y &= 4\end{aligned}$$

They can be written as

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \quad x = \begin{bmatrix} x \\ y \end{bmatrix} \quad b = \begin{bmatrix} 5 \\ 4 \end{bmatrix} \quad Ax = b$$

- A is the coefficient matrix.
- x is the vector of unknowns.
- b is the right-hand-side vector.
- $[A \mid b]$ is the augmented matrix, which attaches b as the last column.

If A has m rows and n columns, then A maps vectors in $\mathbb{R}^n$ to vectors in $\mathbb{R}^m$. Matrix-vector multiplication is defined because the n entries of x match the n columns of A .

Matrix Operations That Matter

For matrices of the same shape, addition is entry by entry. Multiplication by a scalar scales every entry. Matrix multiplication is different: if A is m by n and B is n by p , then AB is m by p , with

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

The order matters. In general, AB is not equal to BA , and one product may exist when the other does not.

The transpose A^T exchanges rows and columns. Useful rules include

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$

The reversal in the last rule is important.

Elementary Row Operations

Three operations preserve the solution set of a linear system:

- 1 Swap two rows.
- 2 Multiply a row by a nonzero scalar.
- 3 Add a multiple of one row to another row.

These operations replace equations with equivalent equations. Applied systematically, they produce an echelon form.

In row echelon form (REF):

- every nonzero row is above every zero row;
- each leading entry, or pivot, is to the right of the pivot above it;
- entries below each pivot are zero.

Reduced row echelon form (RREF) adds two conditions:

- every pivot equals 1;
- every pivot is the only nonzero entry in its column.

REF is usually enough for back substitution. RREF makes the solution structure immediately visible and is unique for a given matrix.

Worked Example: One Solution

Solve

$$\begin{aligned}x + 2y &= 5 \\ 3x - y &= 4.\end{aligned}$$

Start with the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 4 \end{array} \right]$$

Apply $R_2 \leftarrow R_2 - 3R_1$:

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -7 & -11 \end{array} \right]$$

The second row gives $y = 11/7$. Substituting into the first equation gives

$$x = 5 - 2(11 / 7) = 13 / 7$$

So the unique solution is $x = 13/7$, $y = 11/7$. A direct check gives both original equations exactly.

Rank and Consistency

The rank of a matrix is the number of pivot positions in an echelon form. It measures how many independent row or column directions the matrix contains.

For a system with n unknowns:

- If $\text{rank}(A) < \text{rank}([A \mid b])$, the system is inconsistent and has no solution.
- If $\text{rank}(A) = \text{rank}([A \mid b]) = n$, the system has exactly one solution.
- If $\text{rank}(A) = \text{rank}([A \mid b]) < n$, the system has infinitely many solutions, with $n - \text{rank}(A)$ free variables.

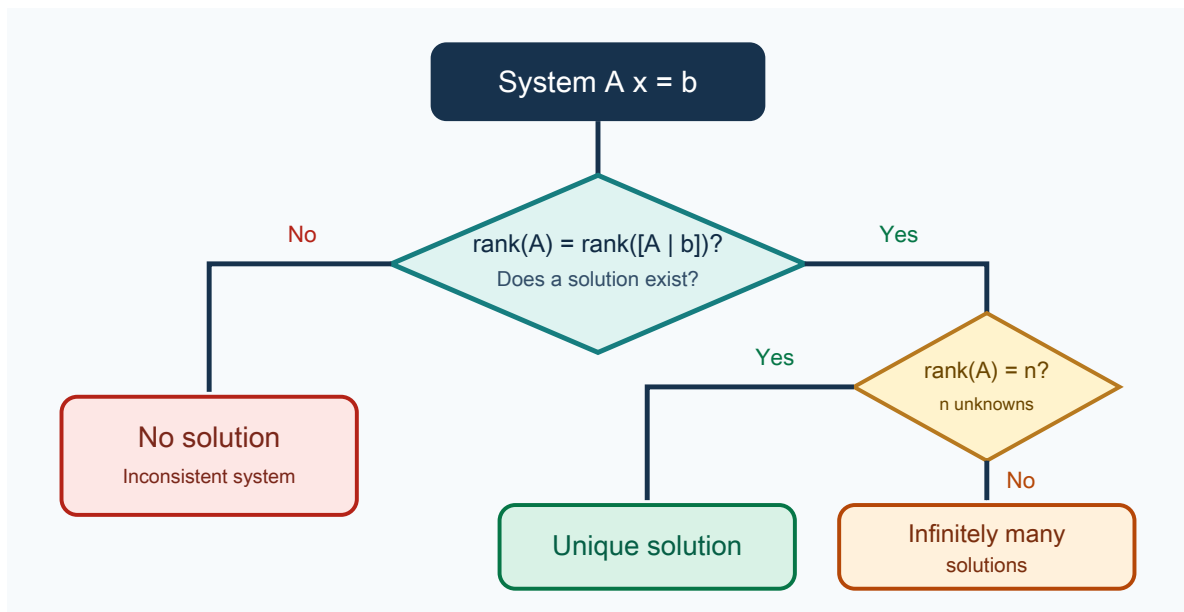


Figure 1. Rank separates inconsistent, uniquely solvable, and underdetermined consistent systems.

Seeing the Three Outcomes

No Solution

$x + y = 1$ and $x + y = 3$ have the same left side but different right sides. Elimination produces a row of the form

$$\begin{aligned} 0x + 0y &= 2 \\ \Rightarrow 0 &= 2 \end{aligned}$$

This contradiction makes the system inconsistent.

Infinitely Many Solutions

$x + y = 2$ and $2x + 2y = 4$ describe the same line. One row reduces to zero, leaving one pivot and one free variable. Writing $y = t$ gives

$$x = 2 - t, \quad \text{and} \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Unique Solution

Two independent equations in two unknowns produce two pivots. Geometrically, two nonparallel lines meet at one point.

Homogeneous Systems

A homogeneous system has the form $Ax = 0$. It always has the trivial solution $x = 0$.

If A has fewer pivots than unknowns, at least one free variable exists and the system also has nontrivial solutions. Therefore,

$$Ax = 0 \text{ has a nontrivial solution} \\ \Leftrightarrow \text{rank}(A) < n$$

The complete solution set of a homogeneous system is the null space of A , a subspace discussed in [Know More](#).

Numerical Practice

For a square, full-rank matrix, numerical software can solve $Ax = b$ directly. Forming A^{-1} first is usually unnecessary and can be less accurate and more expensive.

The condition number estimates sensitivity: a large condition number means small input or rounding changes can produce much larger changes in the computed solution. This is distinct from exact singularity. A matrix may be invertible in exact arithmetic yet numerically difficult to use.

Current-version note: NumPy 2.5 documents `numpy.linalg.solve` for square, full-rank systems. Non-square or rank-deficient systems require a method suited to least squares or underdetermined equations. Numerical rank is tolerance-dependent because floating-point singular values are rarely exactly zero.

Correction: Singular Does Not Always Mean No Solution

The lecture material sometimes compresses the inverse discussion into “determinant zero means no solution.” The technically correct statement is narrower:

- A square matrix with determinant zero has no inverse.
- For a particular $Ax = b$, a singular matrix may give no solution or infinitely many solutions.
- The outcome depends on whether b belongs to the column space of A .

Common Mistakes

- Multiplying matrices entry by entry instead of using row-by-column products.
- Forgetting that $(AB)^T = B^T A^T$ reverses order.

- Applying an operation to A but not to the matching entry in b .
- Counting nonzero entries instead of pivots when finding rank.
- Declaring a system inconsistent merely because it has fewer pivots than unknowns.
- Treating a nearly singular floating-point matrix as if exact symbolic rules guaranteed a stable answer.

Takeaway

Row reduction reveals the structure hidden in $Ax = b$. Rank first decides whether the equations are compatible; when they are, the number of pivots decides whether the solution is unique or contains free directions.

[Detailed](#) [Know More](#) [External Sources](#)

2. Vector Spaces

A vector space tells us which objects may be added and scaled without leaving the world we are studying. This matters in machine learning because feature vectors, parameter vectors, images stored as arrays, and solution sets of homogeneous equations can all be treated with the same rules.

The available lecture evidence supports algebraic structures, fields, vector-space axioms, examples, subspaces, subspace tests, and rank. Linear independence, basis construction, and affine spaces are listed in the handout but are not yet included because substantive lecture evidence is unavailable.

[Detailed](#) [External Sources](#)

Operations and Scalar Fields

A binary operation combines two elements of a set and returns another element of the same set. Addition on real numbers is one example.

A group is a set with one binary operation satisfying closure, associativity, an identity element, and an inverse for every element. If the operation is also commutative, it is an abelian group.

A field supplies the scalars used to scale vectors. The real numbers $\mathbb{R}$ are the usual field in this course. The field rules make addition, subtraction, multiplication, and division by nonzero values behave consistently.

What Is a Vector Space?

A vector space V over a field F has two operations:

- vector addition: $u + v$;
- scalar multiplication: $c u$, where c is in F .

For all vectors u, v, w in V and scalars c, d in F , these operations obey:

- 1 $u + v$ is in V .
- 2 $c u$ is in V .
- 3 $(u + v) + w = u + (v + w)$.
- 4 There is a zero vector 0 with $u + 0 = u$.
- 5 Every u has an additive inverse $-u$ with $u + (-u) = 0$.

- 6 $u + v = v + u.$
- 7 $c(u + v) = cu + cv.$
- 8 $(c + d)u = cu + du.$
- 9 $c(du) = (cd)u.$
- 10 $1u = u$, where 1 is the multiplicative identity in F .

Rules 1 and 3-6 say that vectors form an abelian group under addition. Rule 2 and the remaining rules describe how field scalars act within the space and interact with vector addition.

Validation note: A previous generated version stated closure under addition but omitted the parallel closure-under-scalar-multiplication item. When closure is listed explicitly, both conditions are required; the corrected list includes both.

Examples

Coordinate Vectors

$\mathbb{R}^n$ is the set of all ordered lists of n real numbers. Addition and scalar multiplication happen component by component. It is the standard finite-dimensional vector space used for numeric features.

For example, if daily sales across three regions are represented by $u = [12, 8, 5]^T$ and $v = [3, 6, 4]^T$, then $u + v = [15, 14, 9]^T$ is another vector in $\mathbb{R}^3$.

Matrices

The set of all m by n real matrices is also a vector space. Adding two matrices of the same shape or scaling every entry preserves that shape.

This matters when a dataset itself is treated as one point in a matrix space, such as a grayscale image represented by a fixed-size pixel matrix.

Subspaces

A subspace U of V is a subset that is itself a vector space using the same operations as V .

Most vector-space axioms are inherited automatically. A practical test is enough:

- 1 0 belongs to U .
- 2 If u and v belong to U , then $u + v$ belongs to U .
- 3 If u belongs to U and c is a scalar, then cu belongs to U .

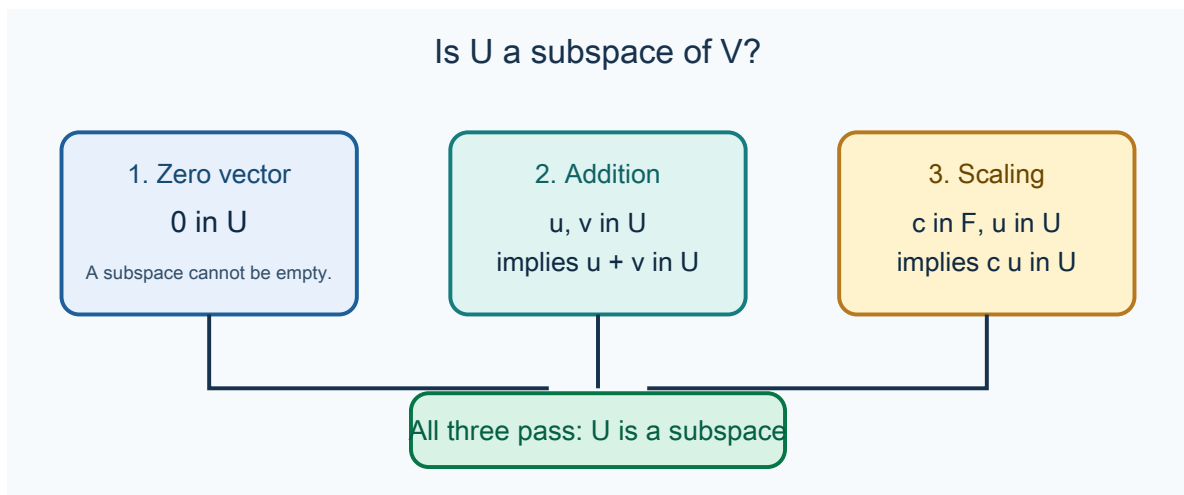


Figure 1. A candidate subspace must contain zero and be closed under addition and scaling.

The last two checks can be combined: for all scalars a, b and all u, v in U , the vector $au + bv$ must also belong to U .

Worked Subspace Check

Let

$$U = \{[x, y, z]^T \in \mathbb{R}^3 : x + y + z = 0\}$$

Zero: $[0, 0, 0]^T$ satisfies the equation.

Addition: If $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$, then

$$(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = 0$$

Scaling: If $x + y + z = 0$, then for any real c ,

$$cx + cy + cz = c(x + y + z) = 0$$

All three checks pass, so U is a subspace of $\mathbb{R}^3$.

Now compare

$$S = \{[x, y]^T \in \mathbb{R}^2 : y = 2\}$$

The zero vector is not in S , so S is not a subspace. Geometrically, it is a line shifted away from the origin.

Important Subspaces of a Matrix

For an m by n matrix A :

- the column space contains all vectors Ax and lies in $\mathbb{R}^m$;
- the null space contains all x satisfying $Ax = 0$ and lies in $\mathbb{R}^n$.

Both are subspaces. The equation $Ax = b$ is consistent exactly when b lies in the column space of A .

Rank as Dimension Information

The rank of A is the number of pivots in an echelon form. It is also the dimension of the column space and the row space.

For an m by n matrix,

$$\text{rank}(A) \leq \min(m, n)$$

Elementary row operations preserve rank. Transposition also preserves rank:

$$\text{rank}(A^T) = \text{rank}(A)$$

In a homogeneous system, the number of free variables is

$$n - \text{rank}(A)$$

This quantity is the dimension of the null space. It explains why fewer pivots than unknowns produce nontrivial homogeneous solutions.

Validation Note: Exact Rank and Numerical Rank

On paper, rank is an exact algebraic property. In floating-point computation, very small singular values must be compared with a tolerance. Therefore, numerical software may classify a nearly dependent direction as zero even when an exact symbolic matrix has full rank.

This is not a contradiction: the numerical question is whether a direction is distinguishable at the available precision.

Common Mistakes

- Calling any subset of $\mathbb{R}^n$ a subspace.
- Checking closure under addition but forgetting scalar multiplication.
- Forgetting that every subspace must contain the zero vector.
- Assuming a shifted line or plane is a subspace even when it misses the origin.
- Confusing the column space, which lies in $\mathbb{R}^m$, with the null space, which lies in $\mathbb{R}^n$.
- Treating a numerical rank result as independent of tolerance.

Takeaway

A vector space is not defined by arrows or coordinates; it is defined by operations and rules. Subspaces preserve those rules inside a smaller set, and matrix rank records how many independent directions survive.

[Detailed Know More External Sources](#)

4. Matrix Decomposition - I

The determinant compresses one important property of a square matrix into a single number. It tells us whether a linear transformation collapses a dimension, how oriented area or volume is scaled, and whether a square system can have a unique solution for every right-hand side.

The available lecture evidence supports only the determinant part of Handout Topic 4. Trace, eigenvalues and eigenvectors, and Cholesky decomposition remain future scope. [Detailed External Sources](#)

Determinant of a 2 by 2 Matrix

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is

$$\det(A) = ad - bc$$

Example:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

gives

$$\det(A) = 3(2) - 1(1) = 5$$

Because the determinant is nonzero, A is invertible.

Minors, Cofactors, and Expansion

For a square matrix, the minor M_{ij} is the determinant left after deleting row i and column j .

The corresponding cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The signs alternate in a checkerboard pattern:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Expanding a 3 by 3 matrix along the first row gives

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

The same determinant results from expansion along any row or column. A row or column containing many zeros is usually the easiest choice.

Worked 3 by 3 Example

Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ -1 & 0 & 3 \end{bmatrix}$$

Expand along the first row:

$$\det(A) = 1 \det \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} + 1 \det \begin{bmatrix} 0 & 2 \\ -1 & 0 \end{bmatrix}$$

Therefore,

$$\det(A) = 1(6) + 1(2) = 8$$

A direct row-reduction or another cofactor expansion gives the same result.

Geometric Meaning

In two dimensions, the columns of A form the sides of a parallelogram. Its area is $|\det(A)|$ when the original unit square has area 1.

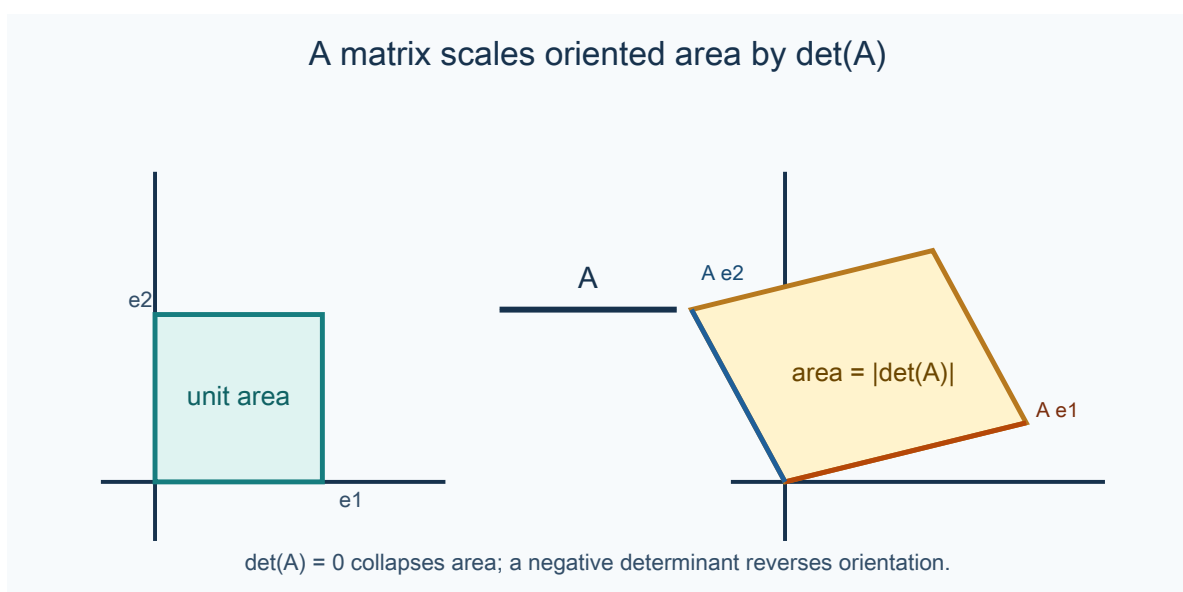


Figure 1. The determinant is a signed area or volume scale factor.

- $|\det(A)| > 1$ expands area or volume.
- $0 < |\det(A)| < 1$ contracts it.
- $\det(A) = 0$ collapses at least one dimension.
- $\det(A) < 0$ reverses orientation in addition to scaling.

This geometric view explains why a zero determinant means the columns cannot span the full space.

Determinant Properties

For square matrices of the same size:

- $\det(AB) = \det(A)\det(B)$.
- $\det(A^T) = \det(A)$.
- Swapping two rows changes the determinant's sign.
- Multiplying one row by c multiplies the determinant by c .
- Adding a multiple of one row to another row leaves the determinant unchanged.
- A triangular matrix has determinant equal to the product of its diagonal entries.
- Equal or linearly dependent rows force the determinant to zero.

These rules often make elimination faster than cofactor expansion for larger matrices.

Determinant and Invertibility

For a square matrix A , the following statements are equivalent:

- $\det(A) \neq 0$;
- A is invertible;
- $\text{rank}(A)$ equals the number of rows;
- $Ax = 0$ has only the trivial solution;
- $Ax = b$ has a unique solution for every compatible-size vector b .

For a 2 by 2 matrix,

$$A^{-1} = \frac{1}{\det(A)} \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

when $\det(A) \neq 0$.

This formula is useful for understanding, but solving a numerical system directly is usually preferable to explicitly forming an inverse.

Correction: Scope of the Inverse Claim

The determinant is defined here only for square matrices. A rectangular matrix has no ordinary determinant, even though it can still have rank and can define a meaningful linear system.

Also, $\det(A) = 0$ means a square matrix is noninvertible. It does not mean every associated system has no solution. Depending on b , the system can be inconsistent or can have infinitely many solutions.

Numerical Caution

Raw determinant magnitude alone cannot diagnose ill-conditioning because rescaling a matrix changes its determinant. For numerical work, inspect the condition number or singular values instead.

Computing a large determinant by recursive cofactor expansion is expensive. Stable numerical libraries use factorization-based methods.

Common Mistakes

- Using $ad + bc$ instead of $ad - bc$ for a 2 by 2 determinant.
- Forgetting the alternating cofactor signs.
- Applying a row operation without accounting for its determinant effect.
- Treating determinant as an entrywise operation.
- Speaking of the determinant of a rectangular matrix.
- Concluding “no solution” from singularity without checking the augmented system.

Takeaway

The determinant connects algebra and geometry. It detects whether a square transformation preserves full dimension, measures signed scale, and links directly to invertibility and unique solvability.

[Detailed](#) [Know More](#) [External Sources](#)